

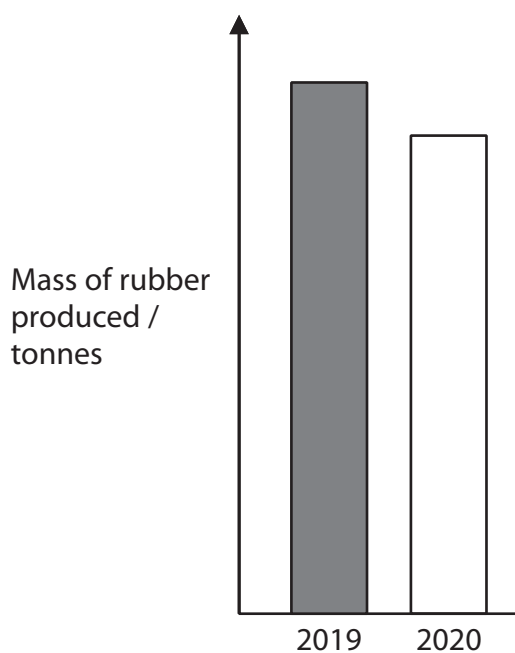
4 Natural rubber is harvested from rubber trees.

The photograph shows rubber being collected from the trunk of a rubber tree in Thailand.



(Source: © Tong Patong/Alamy Stock Photo)

(a) The graph shows the mass of rubber produced in Thailand in 2019 and 2020.



In 2020, Thailand produced 4.37 million tonnes of rubber.

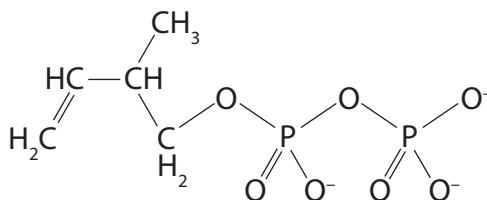
Estimate the mass of rubber produced in Thailand in 2019.

(1)

Answer..... tonnes

- (b) Rubber is synthesised by specialised cells in the trunks of rubber trees.

The diagram shows the structure of one molecule that these cells use to synthesise rubber.



GAP is produced in the light-independent reactions in the leaves.

Explain how the cells in the trunk are able to synthesise the molecule in the diagram using GAP produced in the leaves.

Use the information in the diagram and your own knowledge to support your answer.

(4)

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(c) A rubber tree can be used to harvest rubber until it is about 32 years old.

Explain how dendrochronology could be used to determine the age of a tree that is still producing rubber.

(3)

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(Total for Question 4 = 8 marks)

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